Hospital Healthcare Management: Findings Report

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A review of 2,506 patient records from 2007 across 10 departments: who the patients are, where they were treated, how they fared, how they rated their care and what it cost.

Summary

The hospital treated 2,506 patients in 2007 at a total treatment cost of 404.5K (161 per patient on average) with a mean stay of 1.7 days. Eight findings frame everything that follows:

1. **Cost follows volume.** Average cost per patient is almost identical across the four busiest departments (153 to 158) and across patient types (165 inpatient, 154 outpatient).
2. **Three departments dominate.** OPD, Rehabilitation and Surgery see 71% of patients and generate 67% of cost.
3. **March is the peak.** It has 904 patients (36% of the year) and 147K of cost (36%).
4. **Outcomes are department specific.** All 86 deaths occurred in Rehabilitation (52) and Operation Theatre (34); 80% of readmissions are in Rehabilitation; 190 of the 209 discharges are in Surgery.
5. **Satisfaction is high.** 96% of feedback is positive and the average rating is 4.75 out of 5; Oncology (60% positive, rating 3.73) and Neurology (75%, 4.13) are lowest.
6. **Cost is concentrated.** 190 high-cost treatments (above 500) are 8% of patients but 38% of total cost.
7. **Patient age splits by setting.** Children under 6 and adults over 60 are almost all inpatients, while 21 to 40 is evenly split.
8. **Waiting differs by setting.** Average ER time is 96 for outpatients against 41 for inpatients, about 2.4 times longer.

Overall picture

Activity is dominated by a single month: March accounts for 36% of the year's patients and cost, and every other month stays between 88 and 285 patients.

Month (2007)	Patients	Cost (K)	Avg cost per patient	Inpatient share
January	148	27.0	182	67%
February	123	17.5	143	92%
March	904	147.3	163	57%
April	141	22.9	162	74%
May	200	30.5	153	74%
June	285	50.0	175	66%
July	132	19.2	145	72%
August	201	32.6	162	87%
September	97	14.6	150	77%
October	88	11.8	135	74%
November	99	18.3	185	68%
December	88	12.7	145	36%

- **Volume sets monthly cost.** Average cost per patient moves in a narrow band (135 to 185), so months with more patients cost more almost one for one.
- **Peaks and troughs.** After March, June is the next peak (285 patients, 50K); October and December are the quietest (88 patients each).
- **Patient mix.** 1,679 patients (67%) are inpatients and 827 (33%) are outpatients. December has the lowest inpatient share (36%), February the highest (92%).
- **Length of stay.** 755 patients (30%) stay 0 days, 1,090 (43%) stay 2 days and 267 inpatients stay 3 days or more, up to a maximum of 32.
- **Outcome mix.** 65.3% have a normal outcome, 19.7% are readmitted, 8.3% are discharged, 3.4% die and 3.3% go to ICU.

Patients

Patients are young (average age 30.5, oldest 98) and almost evenly split by gender, and the care setting depends strongly on age.

Age group	Patients	Share	Inpatients	Outpatients	Total cost (K)	Avg cost per patient
0 to 6	317	12.7%	291	26	51.9	164
6 to 20	794	31.7%	480	314	136.0	171
21 to 40	512	20.4%	257	255	76.9	150
41 to 60	695	27.7%	464	231	110.3	159
60+	188	7.5%	187	1	29.4	157

- **Age.** The 6 to 20 group is the largest (794 patients, 32%) and also the costliest group (136K, 34% of total cost).
- **Setting by age.** 92% of children under 6 (291 of 317) and 99.5% of patients over 60 (187 of 188) are inpatients, against 60% for ages 6 to 20 and 50% for ages 21 to 40.
- **Gender.** 1,280 patients are female (51.1%) and 1,226 male (48.9%). Cost is nearly identical: 205.9K for women and 198.5K for men, about 161 to 162 per patient.
- **Geography.** Patients come from 37 states and regions. British Columbia alone supplies 977 (39%), followed by California (402), Washington (215), England (175) and New South Wales (130); these five account for 1,899 patients (76%).
- **Rating by setting.** Outpatients rate their care slightly higher (4.81) than inpatients (4.72).

Departments

Three departments (OPD, Rehabilitation and Surgery) see 71% of patients and generate 67% of cost, while the smallest departments cost the most per patient.

Department	Patients	Total cost (K)	Share of cost	Avg cost per patient	Median cost	Avg stay (days)
OPD	824	126.2	31.2%	153	44.7	0.1
Rehabilitation	515	79.5	19.7%	154	44.7	2.1
Surgery	436	66.9	16.5%	153	43.7	2.8
Operation Theatre	242	38.1	9.4%	158	44.7	2.0
Dermatology	169	31.8	7.9%	188	44.7	2.9
Orthopaedics	96	15.8	3.9%	165	47.4	2.2
Neurology	72	11.6	2.9%	161	62.1	3.8

Department	Patients	Total cost (K)	Share of cost	Avg cost per patient	Median cost	Avg stay (days)
Gynaecology	59	12.7	3.1%	215	152.4	3.9
Cardiology	48	12.8	3.2%	266	50.6	2.0
Oncology	45	9.2	2.3%	205	73.6	2.0
All departments	**2,506**	**404.5**	**100%**	**161**	**44.7**	**1.7**

- **Concentration.** OPD, Rehabilitation and Surgery together treat 1,775 patients and account for 272.5K of cost.
- **Cost per patient rises as volume falls.** Cardiology (266), Gynaecology (215) and Oncology (205) are the costliest per patient but see 59 patients or fewer.
- **What drives the averages.** Cardiology's median is only 50.6 against a mean of 266, so a few very expensive cases lift it. Gynaecology is expensive throughout, with a median of 152, more than three times that of most departments.
- **Length of stay.** OPD averages 0.1 days. Gynaecology (3.9) and Neurology (3.8) have the longest stays, followed by Dermatology (2.9) and Surgery (2.8).
- **Staff.** All 262 staff members treated patients in each of OPD, Rehabilitation and Surgery, so staff work across departments rather than belonging to one.

Outcomes and Experience

Every readmission and death sits in a few departments, and satisfaction is high everywhere except Oncology and Neurology.

Clinical outcomes

Department	Normal	Readmit	Discharge	ICU	Death
OPD	756	66	1	1	0
Rehabilitation	70	393	0	0	52
Surgery	182	22	190	42	0
Operation Theatre	196	12	0	0	34
Dermatology	160	0	0	9	0
Orthopaedics	75	0	18	3	0
Neurology	66	0	0	6	0
Gynaecology	51	0	0	8	0

Department	Normal	Readmit	Discharge	ICU	Death
Cardiology	40	0	0	8	0
Oncology	40	0	0	5	0
All departments	**1,636**	**493**	**209**	**82**	**86**

- **Readmissions.** 493 patients (19.7%) were readmitted: 393 in Rehabilitation (a 76% readmission rate for that department), 66 in OPD (8%), 22 in Surgery (5%) and 12 in Operation Theatre (5%). No other department has any.
- **Deaths.** 86 patients died (3.4%): 52 in Rehabilitation (10% of its patients) and 34 in Operation Theatre (14%). All were aged 41 or older (44 aged 41 to 60 and 42 aged over 60).
- **ICU.** 82 patients (3.3%) went to ICU, 42 of them from Surgery; every department except Rehabilitation and Operation Theatre has some.
- **Discharges.** 190 of the 209 discharges (91%) are in Surgery, 18 in Orthopaedics and 1 in OPD.
- **Stay by outcome.** ICU patients stay longest (4.1 days), then discharged (3.4), deaths (2.0), readmissions (1.8) and normal outcomes (1.3).

Patient satisfaction

Department	Positive feedback	Avg rating (of 5)
Surgery	100%	4.98
Gynaecology	100%	4.97
Orthopaedics	100%	4.94
Dermatology	97.6%	4.89
OPD	96.8%	4.81
Rehabilitation	92.2%	4.66
Cardiology	91.7%	4.60
Operation Theatre	100%	4.50
Neurology	75.0%	4.13
Oncology	60.0%	3.73

- **Overall.** 2,396 of 2,506 responses (95.6%) are positive: 1,828 Fully Agree (73%) and 568 Rather Agree (23%). 81 are negative (3.2%) and 29 are Don't know (1.2%). The average rating is 4.75 out of 5.
- **Lowest satisfaction.** Oncology has 18 Rather Disagree responses out of 45 (40%) and Neurology 18 negative out of 72 (25%).
- **Operation Theatre.** All 242 responses are Rather Agree, with none at Fully Agree, which matches its lower rating of 4.50.
- **Strongest disagreement.** Of the 30 Fully Disagree responses, 21 are in OPD, 4 in Cardiology, 3 in Dermatology and 2 in Neurology.
- **Don't know.** All 29 come from Rehabilitation.

Treatment Costs

Total cost is 404.5K, but a typical treatment is small and a few expensive ones carry the total: the median is 44.7 against a mean of 161, and the highest single treatment is 1,241.

Patient group	Patients	Total cost (K)	Share of cost	Avg cost per patient
Inpatients	1,679	277.4	68.6%	165
Outpatients	827	127.1	31.4%	154
Normal outcome	1,636	264.0	65.3%	161
Readmit	493	80.2	19.8%	163
Discharge	209	31.1	7.7%	149
ICU	82	19.9	4.9%	242
Death	86	9.3	2.3%	108

- **Concentration.** 190 treatments above 500 (8% of patients) account for 154.2K, 38% of all cost.
- **Inpatients and outpatients.** Inpatients carry 69% of cost simply because they are 67% of patients; the average per patient differs by only 11 (165 against 154).
- **ICU.** ICU is the most expensive outcome per patient (242), 50% above a normal case (161).
- **Readmissions.** They cost 80.2K, one fifth of all spend, and most of it comes from Rehabilitation, which holds 393 of the 493 readmissions.

- **Deaths.** These patients cost the least on average (108), consistent with their short stays (2.0 days).
- **Age and gender.** The 6 to 20 group costs the most (136K, 34%), followed by 41 to 60 (110K, 27%); women and men cost almost the same (205.9K and 198.5K).
- **Month.** March costs 147K (36% of the year); the next highest month, June, costs 50K.

Recommendations

The findings point to six areas for action, in priority order.

1. **Review Rehabilitation discharge and follow-up care.** It has 80% of readmissions (a 76% rate), 52 of the 86 deaths and 80K of readmission cost.
2. **Review Operation Theatre outcomes.** 34 of its 242 patients died (14%), the highest death rate of any department.
3. **Plan capacity for peak months.** March carried 36% of the year's patients and cost; understanding what drove it would help staff and bed planning.
4. **Follow up Oncology and Neurology feedback.** Positive rates of 60% and 75% are well below the 92% to 100% elsewhere; both departments are small, so collect more responses.
5. **Plan beds around long stays.** Gynaecology (3.9 days), Neurology (3.8) and the 267 inpatients staying 3 days or more occupy beds longest.
6. **Examine high-cost cases.** 190 treatments above 500 drive 38% of cost, and they lift Cardiology and Oncology well above their medians.

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